4 BILANGAN KOMPLEKS & TEOREMA DE MOIVRE

Parameter : $a=0, b=2 \rightarrow z_1 = (a+2)+(b+1)i = 2+3i$, $z_2 = (b+1)-(a+2)i = 3-2i$

a. konversi bentuk kutub

$z_1 = 2+3i$:

$r_1 = |z_1| = \sqrt{(2^2+3^2)} = \sqrt{(4+9)} = \sqrt{13} \simeq 3.6056$

$\theta_1 = \arctan(3/2) \approx 0.9828$ rad $\approx 56.3099°$

$z_1 = \sqrt{13} \cdot cis(0.9828) \simeq 3.6056 \cdot cis(56.3099)$

$z_2 = 3-2i$:

$r_2 = |z_2| = \sqrt{(3^2+(-2)^2)} = \sqrt{(9+4)} = \sqrt{13} \approx 3.6056$

$\theta_2 = \arctan(-2/3) = -0.5880$ rad $= -33.6901°$

$z_2 = \sqrt{13} \cdot cis(-0.5880) \simeq 3.6056 \cdot cis(-33.6901°)$

$|z_1| = |z_2| = \sqrt{13} \simeq 3.6056 \rightarrow$ amplitudo kedua fasor sama, hanya fase yg berbeda

b. Operasi Aritmatika kompleks

$z_1 + z_2 = (2+3i)+(3-2i) = 5+i$

$z_1 \times z_2 = (2+3i)(3-2i) = 6-4i+9i-6i^2 = 12+5i$

z_1/z_2 (rasionalisasi) $= \frac{(2+3i)}{(3-2i)} \times \frac{(3-2i)}{(2+3i)}$

$= \frac{(6+4i+9i+6i^2)}{(9+4)}$

$= \frac{(6+13i-6)}{13} = \frac{13i}{13} = i$

$z_1/z_2 = 0+i$ (tepat)

$|z_1 - z_2|$:

$z_1 - z_2 = (2+3i)-(3-2i) = -1+5i$

$|z_1 - z_2| = \sqrt{(1+25)} = \sqrt{26} = 5.0990$

c. De Moivre $(z_1)^4$

$z_1 = \sqrt{13} \cdot cis(0.9828) \rightarrow z_1^4 = (\sqrt{13})^4 \cdot cis(4 \times 0.9828)$

$(\sqrt{13})^4 = 13^2 = 169$

$4\theta_1 = 4 \times 0.9828 = 3.9312$ rad

$\cos(3.9312) \approx -0.7101$ | $\sin(3.9312) \approx -0.7041$

$z_1^4 = 169(-0.7101 + (-0.7041)i) \approx -120.01 - 119.00i$

Verifikasi aljabar langsung $(z_1)^2$

$z_1^2 = (2+3i)^2 = 4+12i+9i^2 = 4+12i-9 = -5+12i$

$z_1^4 = (-5+12i)^2 = 25-120i+144i^2 = 25-120i-144$

$= -119-120i$

De Moivre $\rightarrow -120.01 - 119.00i \approx$ Aljabar $\rightarrow -119-120i$ ✓

selisih ≈ 0 (hanya pembulatan)

d. Akar kubik dari $w = 8 \cdot cis(\theta_0)$

$\theta_0 = 10° = 10 \times \frac{\pi}{180} = \frac{\pi}{18} \approx 0.1745$ rad

$w = 8 \cdot cis(10°)$

Akar kubik : $w_k = \sqrt[3]{8} \cdot cis\left(\frac{(\theta_0 + 360° k)}{3}\right)$ utk $k = 0, 1, 2$

$\sqrt[3]{8} = 2$ $(r = 2)$

$k = 0$

$\theta = \frac{(10° + 0°)}{3} = 3.333°$

$= 0.0582$ rad

$w_0 = 2 \cdot cis(3.333°)$

$= 2(\cos 3.333° + i \cdot \sin 3.333°)$

$\approx 2(0.9983 + 0.0581i)$

$\approx 1.9966 + 0.1163i$

$k = 1$

$\theta = \frac{(10° + 360°)}{3} = 123.333°$

$= 2.1533$ rad

$w_1 = 2 \cdot cis(123.333°)$

$= 2(\cos 123.333 + i \sin(123.333)$

$\approx 2(-0.5490 + 0.8358i)$

$\approx -1.0981 + 1.6716i$

$k = 2$

$\theta = \frac{(10° + 720°)}{3} = 243.333$

$= 4.2483$ rad

$w_2 = 2 \cdot cis(243.333°)$

$= 2(\cos 243.333 + i \sin 243.333)$

$\approx 2(-0.4493 - 0.8934i)$

$\approx -0.8986 - 1.7868i$

f. Interpretasi Bisnis

$\arg(z_1/z_2) = \arg(i) = \pi/2 = 90°$

Dalam konteks bisnis, ini berarti produk unggulan (z_1) memiliki fase penjualan mingguan yg bergeser 90° lbh cepat dr produk pesaing (z_2). Artinya, puncak penjualan produk unggulan terjadi seperempat siklus lbh awal dr produk pesaing. Fourier Transform (FT) yg berbasis bilangan kompleks sangat penting utk analisis deret waktu penjualan krn FT mampu menguraikan sinyal penjualan kompleks mjd komponen frekuensi-frekuensinya. Dengan FT, kita bisa mengidentifikasi pola musiman (mingguan, bulanan, tahunan), efek hari libur, dan tren jangka panjang, scr terpisah. Ini memungkinkan tim data utk membangun model forecasting yg jauh lbh akurat dibandingkan regresi linear biasa.

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